

Project Specifications Report

1. Introduction

Recent advances in deep learning and computer vision have significantly transformed image processing, yet a gap remains between algorithmic capability and user accessibility. While professional tools often require complex manual operations, many fully automated solutions lack the flexibility to handle real-world edge cases. Invisio is an AI-powered, web-based platform designed to bridge this gap by automating advanced enhancement tasks through an intuitive, hybrid interface.

Invisio employs a human-in-the-loop strategy, combining robust deep learning pipelines with structured user control to ensure high-quality results even in complex scenarios. The system integrates features such as automated object removal, facial restoration, and neural style transfer into a unified workflow.

Beyond technical performance, Invisio is built on privacy-first architectural principles. By processing images within a temporary session context and ensuring no persistent storage, the platform aligns with modern data protection standards like GDPR and KVKK, providing a secure and ethical environment for creative image manipulation.

1.1. Description

Invisio is a web-based AI-powered photo editor designed to automate common and advanced image editing operations. Users can upload images and apply various enhancements through an intuitive interface without manual adjustments.

The system integrates multiple deep learning models into a unified editing pipeline and processes images temporarily to ensure data privacy.

Key features of Invisio include:

- Automated object removal (AI inpainting-based deletion of unwanted objects)
- Background removal (automatic foreground segmentation)
- Background replacement (template-based or user-uploaded background integration)
- Facial enhancement (face restoration / super-resolution for clearer portraits)
- Facial beautification (skin smoothing and natural retouching)
- Face feature editing: nose resize (enlarge/shrink)
- Face feature editing: eye resize (enlarge/shrink)
- Face feature editing: mouth/lip shape adjustment
- Hair color changing (automatic hair segmentation + recoloring)
- Pose-to-face integration (embedding a selected pose onto another face / face-to-pose alignment)

- Lighting enhancement (automatic exposure correction)
- Contrast adjustment (AI-based contrast balancing)
- Color correction (white balance and color tone optimization)
- Black-and-white image colorization (automatic colorization)
- Neural style transfer (artistic style application)
- Artistic filter suggestions (intelligent, content-aware filter recommendation)
- Real-time preview (instant before/after visualization)
- Image export and download (high-quality output saving)

1.2. Constraints

The project is developed under several technical, functional, and resource-related constraints that define its scope and limitations.

Project constraints include:

- The system supports only image-based processing (no video or audio editing)
- The system supports only web environments.
- Performance depends on GPU availability for model inference
- Only publicly available datasets are used for academic purposes
- Limited development time and computational resources

1.3. Professional and Ethical Issues

The use of AI in image editing raises important ethical and professional considerations, particularly related to privacy, data handling, and responsible image manipulation.

To address these concerns, Invisio is designed with privacy-first and ethical AI principles.

Key professional and ethical considerations include:

- User images are processed in memory and deleted after processing by default.
- Images and generated content are used for training or shared with the community only if the user explicitly opts in.
- Users may choose to share images they like with the community.
- No image is shared unless the user explicitly selects it.
- AI-generated edits are clearly communicated to users
- Features such as beautification avoid promoting misleading or harmful outcomes
- Compliance with data protection regulations (GDPR and KVKK)
- The system shall not allow the upload, processing, or generation of images containing NSFW (Not Safe For Work) content.

2. Requirements

The system requirements define the functional and non-functional expectations of the project.

Functional requirements:

- Upload images via a web interface
- Apply AI-based image enhancement operations
- Display real-time previews of processed images
- Allow users to download edited images

Non-functional requirements:

- Low-latency image processing
- Secure communication between frontend and backend
- Cross-browser compatibility
- High usability for non-technical users
- No permanent storage of user data

3. References

- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., et al. (2014). *Generative adversarial nets*. NeurIPS.
- Gatys, L. A., Ecker, A. S., & Bethge, M. (2015). *A neural algorithm of artistic style*. arXiv preprint.
- Jing, Y., Yang, Y., Feng, Z., Ye, J., Yu, Y., & Song, M. (2017). *Neural style transfer: A review*. IEEE Trans. on Visualization and Computer Graphics.
- Xu, Y. (2025). *Style transfer review: Traditional machine learning to deep learning*. Information, 16(2).
- Zhang, S., et al. (2025). *Applying deep learning for style transfer in digital art*. Scientific Reports.
- Diamant, N., Zadok, D., Baskin, C., Schwartz, E., & Bronstein, A. M. (2019). *Beholder-GAN: Generation and beautification of facial images with conditioning on their beauty level*. In *Proc. IEEE ICIP*.
- Gan, J., et al. (2024). *Applied research on face image beautification based on FIBGAN*. Electronics.
- Peng, T. (2024). *ISFB-GAN: Interpretable semantic face beautification*. Expert Systems with Applications.
- Ronneberger, O., Fischer, P., & Brox, T. (2015). *U-Net: Convolutional networks for biomedical image segmentation*. MICCAI.
- Ling, H., Kreis, K., Li, D., Kim, S. W., Torralba, A., & Fidler, S. (2021). *EditGAN: High-precision semantic image editing*. arXiv preprint.
- Anwar, S., Tahir, M., Li, C., Mian, A., Khan, F. S., & Muzaffar, A. W. (2020). *Image colorization: A survey and dataset*. arXiv preprint.

- [Author(s)] (2024). *[Survey on image colorization]*. Electronics.
- Gui, J., Sun, Z., Wen, Y., Tao, D., & Ye, J. (2020). *A review on Generative Adversarial Networks: Algorithms, theory, and applications*.
- Shamsolmoali, P., Zareapoor, M., Granger, E., Zhou, H., Wang, R., Celebi, M. E., & Yang, J. (2020). *Image synthesis with adversarial networks: A comprehensive survey and case studies*. arXiv preprint.
- Wu, X., Xu, K., & Hall, P. M. (2017). *A survey of image synthesis and editing with generative adversarial networks*. Tsinghua Sci. & Tech.
- Zhou, X. (2025). *Survey paper bridging the metrics gap in image style transfer*. [Journal].
- Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., & Hochreiter, S. (2017). *GANs trained by a two time-scale update rule converge to a local Nash equilibrium*. NeurIPS.
- [Author(s)] (2025). *A review on generative AI for text-to-image and image-to-image generation*. arXiv preprint.